605.486 – Android Development

Module 14
Sensors

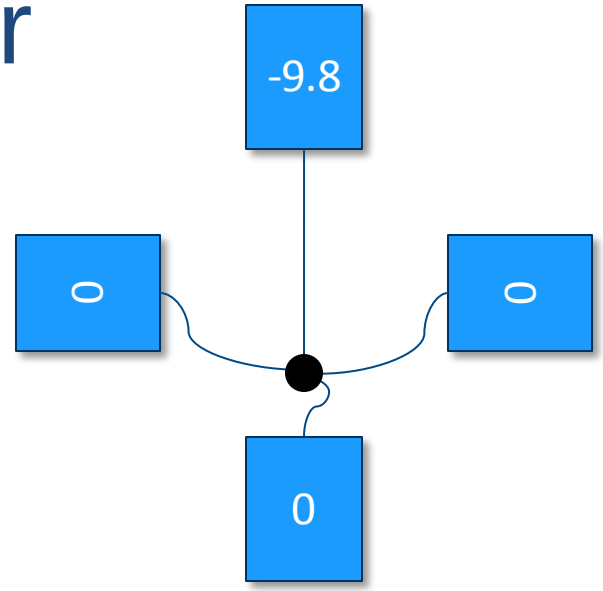
Sensors

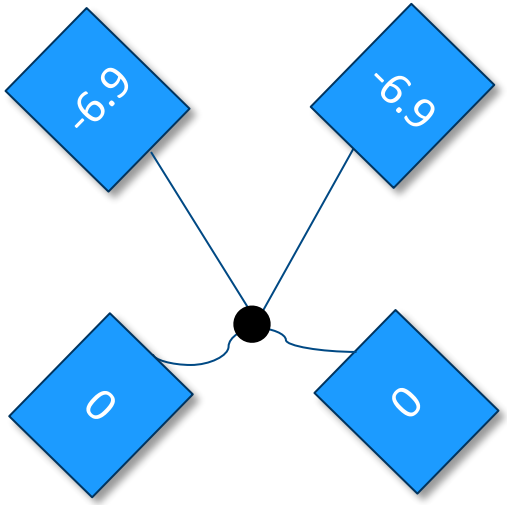
- Each device has different sensors
  - Don't make assumptions!
  - Different quality levels; Often "cheap"

- Common Types
  - Accelerometer
  - Gyroscope
  - Magnetometer
  - Proximity
  - Light
  - Temperature
  - Pressure

Accelerometer

- Easiest to think of in terms of springs/hanging scales
- Measures 3D linear acceleration *including* gravity
- Useful for tilt sensing, changes in speed
- (fictional numbers ->)





Magnetometer

- aka compass
- Three dimensional
  - why does it matter?
- Why might it be noisy?

Sensor Fusion

- Algorithms that combine results of various sensors to refine the results
- See "**Sensor Fusion on Android Devices: A Revolution in Motion Processing**" on youtube
  - David Sachs of InvenSense
  - InvenSense algorithms now incorporated into Android

Virtual Sensors

- Using Sensor Fusion Algorithms

- Gravity
- Linear Acceleration
- Rotation Vector

Example 1 - Which Sensors?

- SensorManager (proxy to Sensor Service)

```
SensorManager sensorManager =
  (SensorManager) getSystemService(SENSOR_SERVICE);
List<Sensor> sensorList =
  sensorManager.getSensorList(Sensor.TYPE_ALL);
```

Example 2 - Accelerometer Values

- Layout to display x,y,z accel values
- SensorListener
- X, Y, Z axes are ALWAYS fixed
  - Y is “up” as you would normally look at device
    - Phone – portrait
    - Tablet – landscape

- Look at SensorEvent javadoc to see what event parameters are available for each sensor type

- Sensor Delays
  - Fastest – get data as fast as possible - avoid
  - Game – suitable for game change rates
  - Normal – good for screen orientation changes
  - UI – general user interface sensor access

Example 3 - Orientation Fixup

- Ask display for rotation
  - 0, 90, 180, 270
  - Pick x,y appropriately

Example 4 - Custom View - Puck

- PuckView (don’t pronounce out loud)
- Similar to bouncing ball but allows x,y accel to change based on accelerometer
- Fix to portrait mode…

Example 5 - Magnetometer

- Measure magnetic fields
- Example shows magnitudes

Example 6 - Camera Preview

- Display image seen from camera
- Uses push buffer – blasts image to screen
- SurfaceView – “hole” through window directly to canvas
- Requires CAMERA permission
- Lock to landscape

Example 7 - Camera Overlay

- Augmented reality
- Draw circle on top of camera image

Example 8 - Draw axis

- Use magnetometer to determine bearing
- Draw x-axis on top

Example 9 - Low-pass filter

- Smoothens changes - slows down amount of change
- Many algorithms – much better ones available

Example 10 - Vertical Axis

- Add y axis
- Possible game

Example 11 - Vibration

- Vibrator is a manager for Vibration Service

```
private long[] vibrationPattern =
 {0, 50, 50, 50, 50, 200, 100, 50};

// start after 0ms
// vibrate for 50ms
// off for 50ms
// vibrate for 50ms
// off for 50ms
// vibrate for 200ms
// off for 200ms
// vibrate for 50 ms

vibrator.vibrate(vibrationPattern, -1);
```